Monitoring Glacier Retreat in Switzerland through Attention U-Net Segmentation

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Deforestation Detection from Satellite Imagery

Amazon Rainforest covers about **40% of Earth's tropical forests**, and serves as a habitat for roughly **10% of all species** worldwide

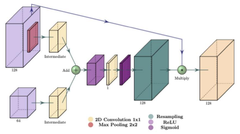
Large-scale deforestation

(about 2300 hectares/day (2020))

A major **threat** to species, a **contributor** to accelerating climate impacts, and a **danger** to the forest's long-term survival

Attention U-Net Model to monitor deforestation from Sentinel-2 satellite imagery

- Encoder: 4 down sampling blocks
- Decoder: 4 up sampling blocks
- Output: Sigmoid activation
- Loss: Binary Cross-Entropy
- 4-band Amazon Forest data model



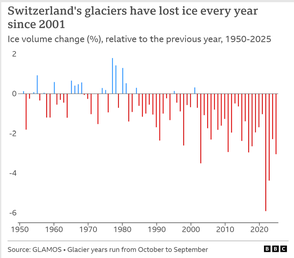
Strong performance:

- Validation set (own replication): F1-Score = 0.9239
- Test set (own replication): F1-Score = 0.8940

Monitoring Glacier Retreat in Switzerland

Because of global warming, **Switzerland's glaciers are melting at an alarming rate:**

- Swiss Alps have **lost 25%** of its volume since 2015
- **Over 1k** small glaciers have already disappeared
- **10% decrease in volume** between 2022 and 2023
- One of **Europe's largest freshwater reservoir**



Unstable **water supply**

Hydropower instability

Increase in **natural hazards**

Impact on **biodiversity**

Tourism, cultural value and global heritage

Monitoring Glacier Retreat:

13 CLIMATE ACTION

Early detection of collapses and unstable slopes ➔ strengthen resilience to climate-related hazards **(13.1)**
Continuous glacier data for hydropower planning ➔ integrate climate measures into national policies **(13.2)**
If transparent monitoring ➔ improved climate **awareness (13.3)**

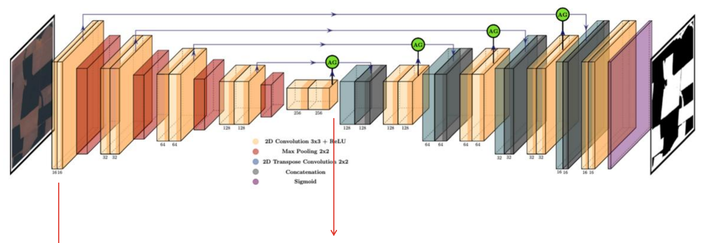
6 CLEAN WATER AND SANITATION

Observations inform LT freshwater availability ➔ **increase water-use efficiency (6.4)**
Safeguard hydrological systems (6.6)

11 SUSTAINABLE CITIES AND COMMUNITIES

Supports **early-warning** for communities downstream ➔ reducing disaster risks and impacts **(11.5)**

Adapted Model Architecture



Add dropout layer (50% dropout rate)

Retrained from scratch to adapt to new input size (512,512, 4) ➔ (512,512, 6)

Loaded pre-trained Amazon weights ➔ full fine-tuning of all layers

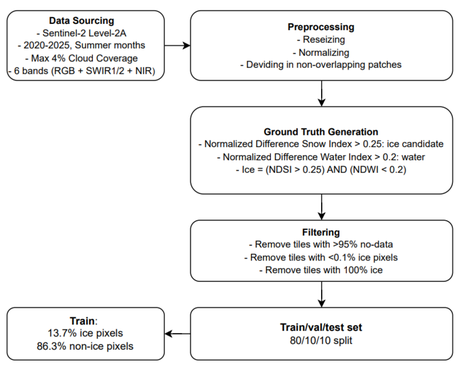
Other adaptations:

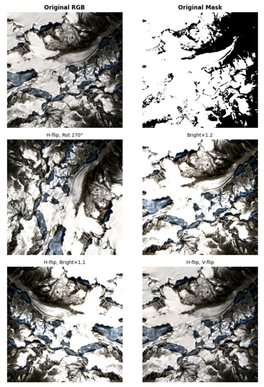
- Learning rate scheduler
- SDG ➔ Dice Loss
- Data Augmentation
- Batch Size
- Early Stopping

Optimal parameters:

```
PARAMETERS = {
    'batch_size': 2,
    'epochs': 60,
    'loss': Dice Loss,
    'learning_rate': 0.0001,
    'reduce_learning_rate_factor': 0.3,
    'patience_learning_rate': 5,
    'drop_rate': 0.5,
    'patience_early_stopping': 10}
```

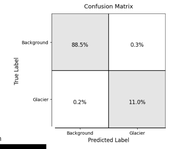
Sentinel-2 satellite images of Swiss Alps (Jungfrau-Aletsch region)

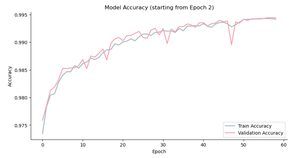


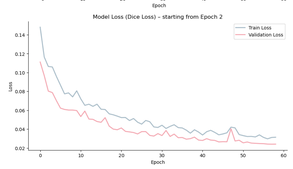


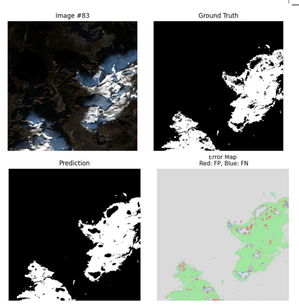
Adapted Model Achieves Stellar Results

Split	IoU	Precision	Recall	F1
Validation	0.9892 [0.9865, 0.9919]	0.9945 [0.9931, 0.9959]	0.9944 [0.9930, 0.9958]	0.9944 [0.9930, 0.9958]
Test	0.9910 [0.9885, 0.9936]	0.9954 [0.9942, 0.9967]	0.9954 [0.9941, 0.9967]	0.9954 [0.9941, 0.9967]

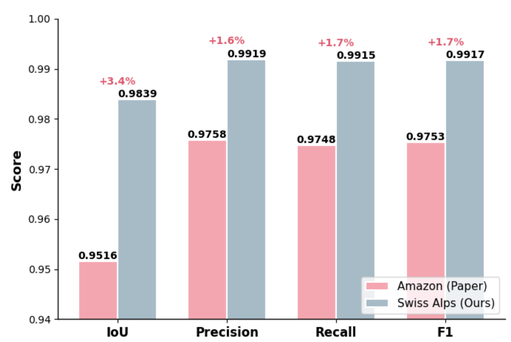


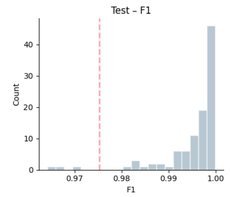


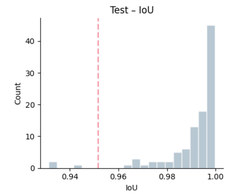




Difference in Performance is Statistically Significant







Discussion and Critical Reflections on SDGs

Why Better Performance?

- Task characteristics: Ice has unique spectral signature
- More spectral info: 6 bands vs 4 bands
- More training data (800 vs. 200) + augmentation
- More advanced model (dropout, LR scheduler)

Sustainability Trade-offs:

- Trained on one region ➔ generalisable?
- No community engagement, who is in control?
- Impact depends on whether **policymakers** and local authorities **act on the generated insights**
- **Communities** downstream need to receive timely accessible information
- **Benefit**: Scalable, frequent monitoring
- **Cost**: Infrastructure dependencies (satellites, GPUs), carbon cost for training

https://github.com/camilletyriard-dev/Deforestation_Detection